

Docker Container Stuck In Restarting SOP

Objective

Stabilize a Docker container that continuously restarts due to application crash, bad configuration, dependency failure, or incorrect restart policy.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
- 2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
- 3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
- 4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
- 5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T036
Issue	Docker container stuck in restarting
Category	Deployment
Service	Docker
Severity	High
Status	Resolved
Description	Container continuously restarting
Expected Resolution	Check logs and fix application crash

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L022	5/1/2026 10:11	ERROR	Container restarting repeatedly	DOCKER_137

Procedure

- 1. Confirm container restart count, current status, exit code, restart policy, and last successful start time.
- 2. Collect current and previous logs to identify the crash reason.
- 3. Inspect entrypoint, command, environment variables, mounts, ports, and health check configuration.
- 4. Validate external dependencies such as database, cache, API, network, and file paths.
- 5. Temporarily disable restart loop only if approved for debugging.
- 6. Fix the root cause such as missing config, dependency outage, permission error, or code crash.
- 7. Recreate container and verify it remains Up without repeated restarts.
- 8. Monitor restart count and application health for at least 15 minutes.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Container restarting repeatedly	Application crash	Check logs and fix crash
Restarts after health check	Health check too strict or wrong path	Tune health check
Dependency unavailable	Database or network issue	Restore dependency or update config
Restart loop hides logs	Aggressive restart policy	Temporarily stop for debugging

Best Practices

1. Monitor restart counts and crash loops.
2. Keep startup dependency checks clear in logs.
3. Use safe health check thresholds.
4. Avoid masking application crashes with restart policies.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook